

Varun Saini

Preston, United Kingdom | 07930421377 | varunsaini1620@gmail.com | github.com/VarunSaini05

PROFESSIONAL SUMMARY

Aerospace engineering specialist with hands-on experience in CFD, ANSYS, Simulation. Proven ability in CFD simulation, propulsion component design, and engineering validation using ANSYS and SolidWorks.

TECHNICAL SKILLS & TOOLS

Engineering: CFD, ANSYS Fluent, ANSYS Mechanical, SolidWorks, Simulation

Programming & Automation: Python, VBA, MATLAB

Data & Analytics: Pandas, Streamlit, ETL Pipelines, Data Analysis

Documentation: Technical Documentation, Manufacturing Workflows, Engineering Reporting

WORK HISTORY

RESEARCH INTERN

University of Lancashire | Preston, United Kingdom

- Performed CFD analysis using ANSYS Fluent for liquid hydrogen propulsion systems.
- Conducted mesh convergence studies and thermo-structural validation workflows.
- Developed MATLAB models supporting propulsion performance analysis.

CAD DESIGNER (2D & 3D)

Kaizen Industries | Remote (Part-Time)

- Generated simulation-ready CAD geometry for aerodynamic and propulsion component analysis using SolidWorks.
- Supported CAD-to-mesh workflows for CFD validation and ANSYS integration.
- Produced technical documentation to enable engineering validation and manufacturing handoff.

LOGISTICS SUPERVISOR

Booths | United Kingdom

- Coordinated logistics support for engineering projects, ensuring parts and materials were available for aerospace validation workflows.
- Documented inventory control procedures and provided technical reporting to support manufacturing handovers.
- Applied data-driven process improvements to improve supply chain reliability for production teams.

VOLUNTEER CITIZEN SCIENTIST

Zooniverse

- Reviewed astronomy datasets to identify supernovae, black holes and deep-space anomalies.

VOLUNTEER CITIZEN SCIENTIST

NASA Globe Mission

- Contributed to environmental and climate observation workflows through scientific data collection and analysis.

CLASS REPRESENTATIVE

University of Lancashire | Preston, United Kingdom

- Represented student engineering cohorts, coordinated academic feedback and organised technical activities.

PROJECTS

Hydrogen Propulsion Nozzle CFD & Automation System

Developed and validated a high-enthalpy hydrogen propulsion nozzle using CFD and thermo-structural ANSYS workflows, focusing on propulsion performance and simulation accuracy.

ANSYS Fluent, ANSYS Mechanical, SolidWorks, MATLAB, Excel

Fusion Reactor DEC System

Developed conceptual Direct Energy Conversion (DEC) models and simulation workflows using MATLAB and SolidWorks for propulsion research.

Python, MATLAB, ANSYS, Simulink, SQL

Aircraft Predictive Maintenance AI System

Designed predictive-maintenance AI workflows for aircraft engine health using NASA CMAPSS datasets to support safety and reliability.

Python, NASA CMAPSS Dataset, Machine Learning

EDUCATION

BEng with Honours in Aerospace Engineering with Pilot Studies (Graduated 2026)

CAD, Structure & FEA, Propulsion, Pilot Studies, Thermo-Fluids, Aeromechanics, Aircraft Design, Project Management

University of Lancashire, United Kingdom

Diploma of Higher Education: Mechanical Engineering (09/2022 – 07/2023)

Electrical and electronic principles, Engineering Mathematics, Strength of Materials, Mechanical Design and Engineering Applications, Mechanical Science, English Literature

PSB Academy, Singapore

German Language Course (08/2020 – 08/2022)

Language Pantheon, Delhi

Certificate of Higher Education: Medical and Engineering (05/2018 – 06/2020)

Himalaya Public Sr. Sec. School, Delhi

CERTIFICATIONS

- NASA: Learning from Dark Matter and Energy
- NASA: Explore Space Tech: Rockets 101
- NASA: Technology Drives Exploration: Detecting Exoplanets
- IARC: Life in Space

PROFESSIONAL MEMBERSHIPS

IMechE, The Institute of Engineers Singapore, NVIDIA Developer Program, JAXA ALOS Global User

LANGUAGES

English (Fluent), Hindi (Native), German (Intermediate)

References available on request